

Math728J – Homework 3

1.
 - a. Given an example of a matrix that is not diagonalizable.
 - b. What can be said of the upper Hessenberg matrix $\mathbf{H}_k$ when $\mathbf{A}$ is symmetric? What does this imply for the Arnoldi process?
 - b. What can be said of the upper Hessenberg matrix $\mathbf{H}_k$ when $\mathbf{A}$ is real skew-symmetric ($\mathbf{A}^T = -\mathbf{A}$)? What does this imply for the Arnoldi process?

2. Let

$$\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{Y}_2 & & & & \\ & \mathbf{I} & \mathbf{Y}_3 & & & \\ & & \cdots & \cdots & & \\ & & & \mathbf{I} & \mathbf{Y}_{k-1} & \\ & & & & \mathbf{I} & \mathbf{Y}_k \\ & & & & & \mathbf{I} \end{pmatrix}$$

- a. Show that $(\mathbf{I} - \mathbf{A})^k = 0$.
 - b. Assume that the GMRES is used to solve a linear system $\mathbf{A}\mathbf{x} = \mathbf{b}$. What is the maximum number of steps that GMRES would require to converge to the exact solution?
3. Let $\mathbf{A} = \mathbf{J}_\lambda$ be an elementary Jordan block of size N corresponding to an eigenvalue $\lambda \neq 0$. This means that

$$\mathbf{A} = \begin{pmatrix} \lambda & 1 & 0 & & \\ 0 & \lambda & 1 & 0 & \\ & \cdots & \cdots & \cdots & \cdots \\ & & 0 & \lambda & 1 & 0 \\ & & & 0 & \lambda & 1 \\ & & & & 0 & \lambda \end{pmatrix}$$

Given an example of $\mathbf{x}_0, \mathbf{b} \in \mathbb{R}^N$ and $\lambda \in \mathbb{R}$ such that the GMRES residuals satisfy $\|\mathbf{r}_k\|_2 = \|\mathbf{r}_0\|_2$ for all $k < N$.

4. (**Programming Exercise**) Let $\Omega = (0, 1)^2$. Consider numerical solution of the boundary value problem

$$\begin{cases} -u_{xx}(x, y) - u_{yy}(x, y) + a1(x, y)u_x(x, y) \\ \quad + a2(x, y)u_y(x, y) + b(x, y)u(x, y) = f(x, y), & (x, y) \in \Omega \\ u(x, y) = 0, & (x, y) \in \partial\Omega \end{cases} \quad (1)$$

where $a1(x, y) = 1$, $a2(x, y) = 10$ and $b(x, y) = 1 + x^2$. We let $u(x, y) = 10.0xy(1 - x)(1 - y)$ and then the right hand side f is determined correspondingly. We consider an uniform square grid with grid points (x_i, y_j) which are chosen to be equidistant:

$$x_i = ih, \quad y_j = jh, \quad i, j = 0, 1, \dots, N \quad (h = 1/N)$$

and the center difference scheme:

$$\begin{aligned} & \frac{-U_{i-1,j} - U_{i,j-1} + 4U_{i,j} - U_{i+1,j} - U_{i,j+1}}{h^2} \\ & + a1_{i,j} \frac{U_{i+1,j} - U_{i-1,j}}{2h} + a2_{i,j} \frac{U_{i,j+1} - U_{i,j-1}}{2h} + b_{i,j} U_{i,j} = f_{i,j}, \\ & i, j = 1, 2, \dots, N-1, \end{aligned}$$

where $a1_{i,j} = a1(x_i, y_j)$, $a2_{i,j} = a2(x_i, y_j)$, $b_{i,j} = b(x_i, y_j)$ and $f_{i,j} = f(x_i, y_j)$, and

$$U_{i,j} \approx u(x_i, y_j), \quad i, j = 1, 2, \dots, N-1$$

$$U_{i,0} = U_{i,N} = U_{0,i} = U_{N,i} = 0, \quad i, j = 0, 1, \dots, N.$$

Let $\mathbf{x}$ be the vector consisting of unknowns $U_{i,j}$ ($i, j = 1, 2, \dots, N-1$) ordered row-wise (from left to right) and upward. Then, we get a linear system such as

$$\mathbf{A}\mathbf{x} = \mathbf{b}$$

where $\mathbf{A} \in \mathbb{R}^{(N-1)^2 \times (N-1)^2}$ is a sparse matrix and $\mathbf{b} \in \mathbb{R}^{(N-1)^2}$. For $N = 10, 20, 40, 80$, determine an approximate solution of (1) using the standard GMRES method (i.e., `gmres()` in our lecture nodes), where the iteration is stopped when either more than $Kmax = 200$ iterations are reached or the residual $\|\mathbf{r}_k\|_2 < \epsilon \|\mathbf{b}\|_2$ with $\epsilon = \frac{h^2}{100}$. Choose $\mathbf{x}_0 = 0$ as the initial value for the iterations. Hand in your codes with some explanations. Provide the number of required iteration steps k , and the errors $\max_{i,j=1,2,\dots,N-1} |U_{i,j} - u(x_i, y_j)|$ in a table. Also plot the history diagram of number of iterations k vs. the relative residual $\|\mathbf{r}_k\|_2 / \|\mathbf{b}\|_2$ in a semi-log scale for the $N = 40$ case using Matlab (the command *semilogy*).

5. **(Programming Exercise)** Re-do Problem 4 using the restarted GMRES method (i.e., `gmresm()` in our lecture nodes) with $m = 5$.